

“Make sure that the kits are expandable so that you can keep developing more and more complex projects.”—Adwait Deshpande, R&D Head, Robotics Division, ThinkLABS, Sine IIT Mumbai



for Microsoft Robotic Studio is also available.

Functions depend on the type of the kit:

- Wheeled robots: Remote control, sensor inputs, line following, motor encoders, LCD, switches, etc
- Walking robots: Motion sequencing, balancing, etc
- UAVs and fully autonomous outdoor robots: GPS navigation and long-distance communications

The modularity of the hardware and user-friendliness of the software encourage users to try out more and more complex contraptions.

Most of the robotic kits are USB-compatible for reprogramming. Hardware structure of certain models can be assembled in multiple ways, so you can construct different robots using the same parts.

Robotic kits are available in a



Fire Bird V ATMEGA2560 hexapod from NEX Robotics

compact form and are easy to interconnect. There will be specification and directions for each and every part, so little effort is required to develop an amazing robot. GUI-based programming software that come along with

driving, servo driving and wireless command are in demand. Most buyers in India, at present, show interest in line-following or maze-solving robots. These robots need reconfigurable maze-solving algorithm. The robot program consists of different algorithms and the user can easily reconfigure the path using the interfacing software. GUIs developed in ROSSUM, XRCL or .NET

with the help of compiler (for micro programming) is used to develop this interface.

There is a high demand for modular robotic kits. A modular robotic kit is a programmable bot that can be interfaced to different motors and



“The level of difficulty is increasing from doing just line-following and obstacle-avoiding.”

—(L) Fahad Azad, Co-founder, Robosoft Systems, and (R) Syed Ross Farooq, Founder, Robosoft Systems



the kit let you develop a robotic application by means of the mouse pointer (requiring only clicks and drags). Usually, robotic kits come with shaped parts of the robots having specific part numbers and associated motors, along with the motor holder. The kit consists

of separate packets of assembly tools and instructions. Software support makes the assembly easy. User manual illustrates assembly step by step with figures. Certain robots need pre-correction of the motors and there are software tools to do this job.

Functionalities in demand

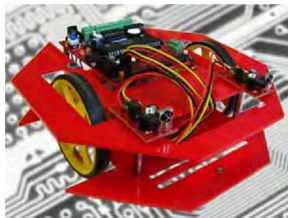
Basic functionalities like sensor interfacing, motor

sensors. It is generally, but not necessarily, a two-wheeled robot with facilities of multiple sensor interface, wireless communication and PC interface.

Level of autonomy

The robots now developed in India use different sensors (such as proximity sensors and polarisation sensors) to maintain the balance of the robot and give direction to its path. Balancing is an amazing process and hence most companies are focused on this function. Various input applications like voice processing, image processing and multiple sensor reading and response are already implemented. But certain materials for manufacturing specialised robots like wall-climbing robots are not available in India, and are very costly to import.

System engineering of industrial



ThinkLABS' iBot microcontroller-based autonomous robotic development platform